Chapter 12: Racing Seconds

Worksheet 12: Racing Seconds

Learning Objective: Measure and compare time in seconds and minutes; solve problems involving the timing of races and sports events.

Worked Example:
A runner completes a 100 m race in 14 seconds, and another in 16 seconds. Who is faster, and by how many seconds?

Solution: The first runner is faster (less time taken), by $16 - 14 = 2$ seconds.

Questions (1–20)

1. How many seconds are there in 1 minute?

2. Convert 2 minutes into seconds.

3. Convert 150 seconds into minutes and seconds.

4. Which is longer: 90 seconds or 1 minute 20 seconds?

5. Write 3 minutes 15 seconds in seconds only.

6. In a 200 m race, Asha finishes in 38 seconds and Priti in 42 seconds. Who wins, and by how many seconds?

7. A swimmer completes one lap in 45 seconds. How long will 4 laps take at the same pace?

8. A cricket bowler takes 25 seconds between each ball. How long will an over of 6 balls take?

9. A relay team of 4 runners each take 50 seconds for their part. Find the total time for the team.

10. A stopwatch shows a runner's time as 1 minute 8 seconds. Convert this into seconds only.

11. In a 400 m race, four runners' timings are 58 sec, 62 sec, 55 sec, 60 sec. Arrange the runners from fastest to slowest.

12. A cyclist completes a 5 km track in 8 min 30 sec on the first attempt and 7 min 50 sec on the second. By how many seconds did the cyclist improve?

13. Heat 1 winner: 13.2 sec; Heat 2 winner: 12.8 sec. Who has the faster time, and what is the difference?

14. A player runs the first lap in 58 sec and the second lap in 1 min 5 sec, against a target of under 1 minute. In which lap did the player meet the target?

15. Four sprinters' times are 11.5 sec, 11.9 sec, 11.2 sec, 11.7 sec. Find the difference between the fastest and slowest time.

16. In a 4×100 m relay, four runners take 12.5 sec, 13.0 sec, 12.8 sec and 12.2 sec for their legs. Find the team's total time.

17. A school sports day record for the 100 m race was 13.4 seconds. This year, a student ran it in 12.9 seconds. By how many seconds was the old record beaten?

18. A marathon checkpoint shows a runner passed at 45 min 20 sec from the start. The runner finishes 3 hours 12 min 40 sec after starting. How much time elapsed between the checkpoint and the finish?

19. Time yourself running or walking 50 steps using a phone or clock, and note your time in seconds. Compare with a friend's time.

20. Recall a sports day or cricket match timing record (race time, bowling time, etc.) and write it down, converting between seconds and minutes if needed.

Reflection Question: Why do you think races are measured in seconds rather than minutes or hours? Write your thoughts.

Real-Life Application Question: Recall watching a cricket or athletics match. Note down one moment where time (seconds or minutes) played an important role, for example the last few seconds of a match.